

Multiple Choice (10 marks)

1. Your first action following confirmation of cardiac arrest is to:
 - (a) ensure the emergency team/services are called.
 - (b) administer an electric shock using a defibrillator.
 - (c) give 30 chest compressions.
 - (d) check the notes to see if the patient has a DNAR order.
 - (e) rush the patient to the emergency room.
2. Which of the following cannot be a substrate for gluconeogenesis?
 - (a) Palmitic acid
 - (b) Pyruvate
 - (c) Galactose
 - (d) Glycerol
 - (e) Propionic acid
3. Life stories are more likely for people who have a sense of
 - (a) Superiority
 - (b) Guilt
 - (c) Personal control
 - (d) Material wealth
 - (e) Humor
4. Which of the following is NOT one of the more frequently named activities by those who are retired?
 - (a) Starting a new business
 - (b) Working part-time
 - (c) Mountain climbing
 - (d) Household chores
 - (e) Travel
5. If the frequency of males affected with an X-linked recessive condition in a human population is .10 (one in ten), what will be the expected frequency of affected females?
 - (a) 0.001
 - (b) 0.01

- (c) 0.1
- (d) 0.15
- (e) 0.0011

True / False (10 marks)

Answer True or False

6. True or False: Vitamin D does not function as a lipid-soluble antioxidant in cell membranes.
7. True or False: Macrostomia results from the failure of fusion of the left and right maxillary processes during embryonic development.
8. True or False: Cerebrospinal fluid circulates between the skull and dura mater surrounding the brain.
9. True or False: Inadequate MHC Class I presentation hinders the effective response of CD 8 T cells in eliminating HIV in chronically infected individuals.
10. True or False: Damage to the third cranial nerve typically results in pupillary dilatation and downward strabismus due to impaired ocular muscle function.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of genetic factors in longevity, alongside mental stimulation, exercise, and diet, and analyze how they collectively contribute to a long and healthy life.
12. Discuss the prevalence of alcoholic myopathy among individuals with alcohol use disorder, analyzing the contributing factors and implications for treatment and recovery.

End of Paper